

APPLICATION LAYER
OS Shell, SDK Games, User Programs

TAL Compiler | Fantasy Console SDK

ASSEMBLER
Two-pass: labels, branch resolution

MACHINE CODE ENCODER/DECODER
27 opcodes, variable-length ternary strings

CPU
Fetch-decode-execute, 27 opcodes, 4 regs

ACCELERATOR | HARDWARE SIM
Tensor, SIMD, GPU | Pipeline, Cache, BP, DMA

ALU | Registers | Memory | Display | Arithmetic

conversion.py | logic.py | adder.py | native C backend